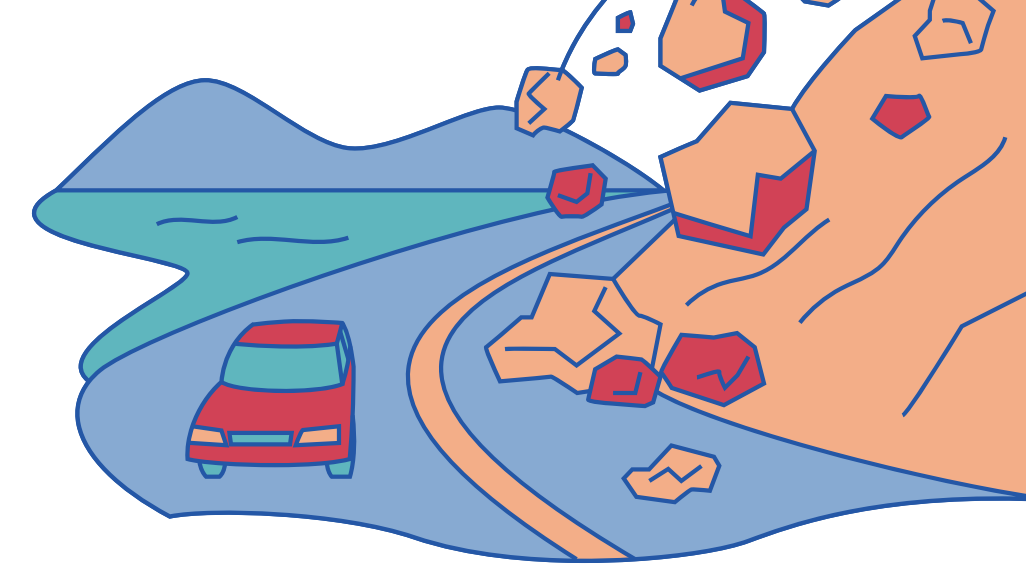


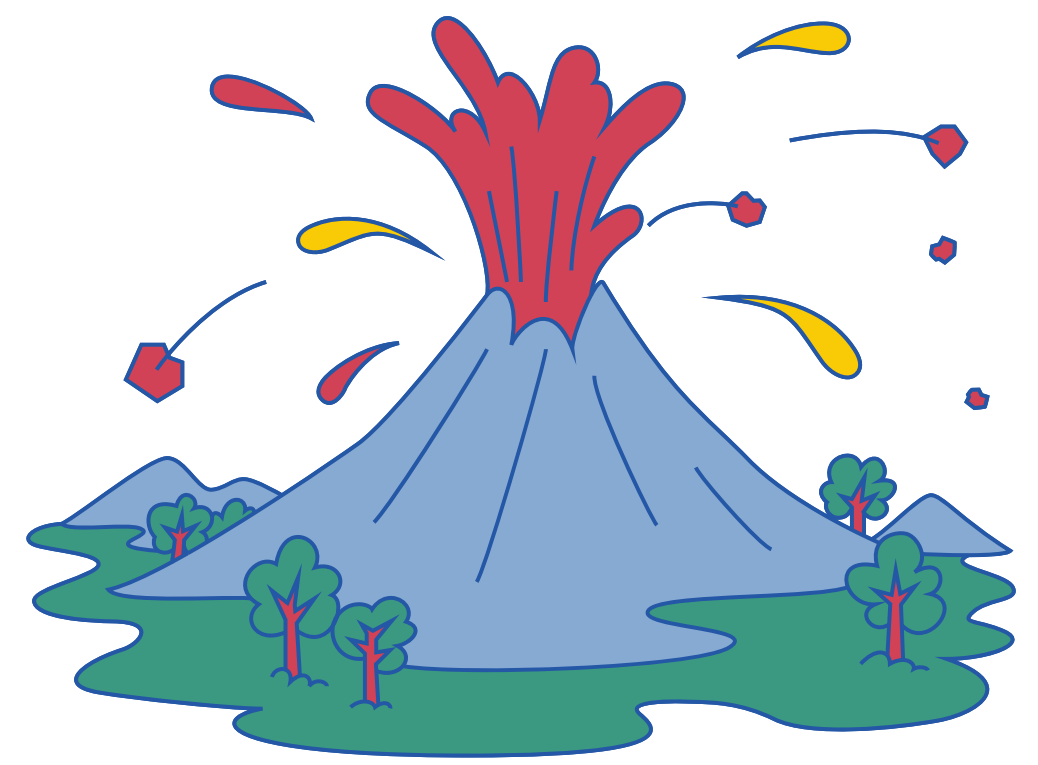


Smart Disaster Prediction and Relief Management System

Presented by Team Pentonic



Team Members



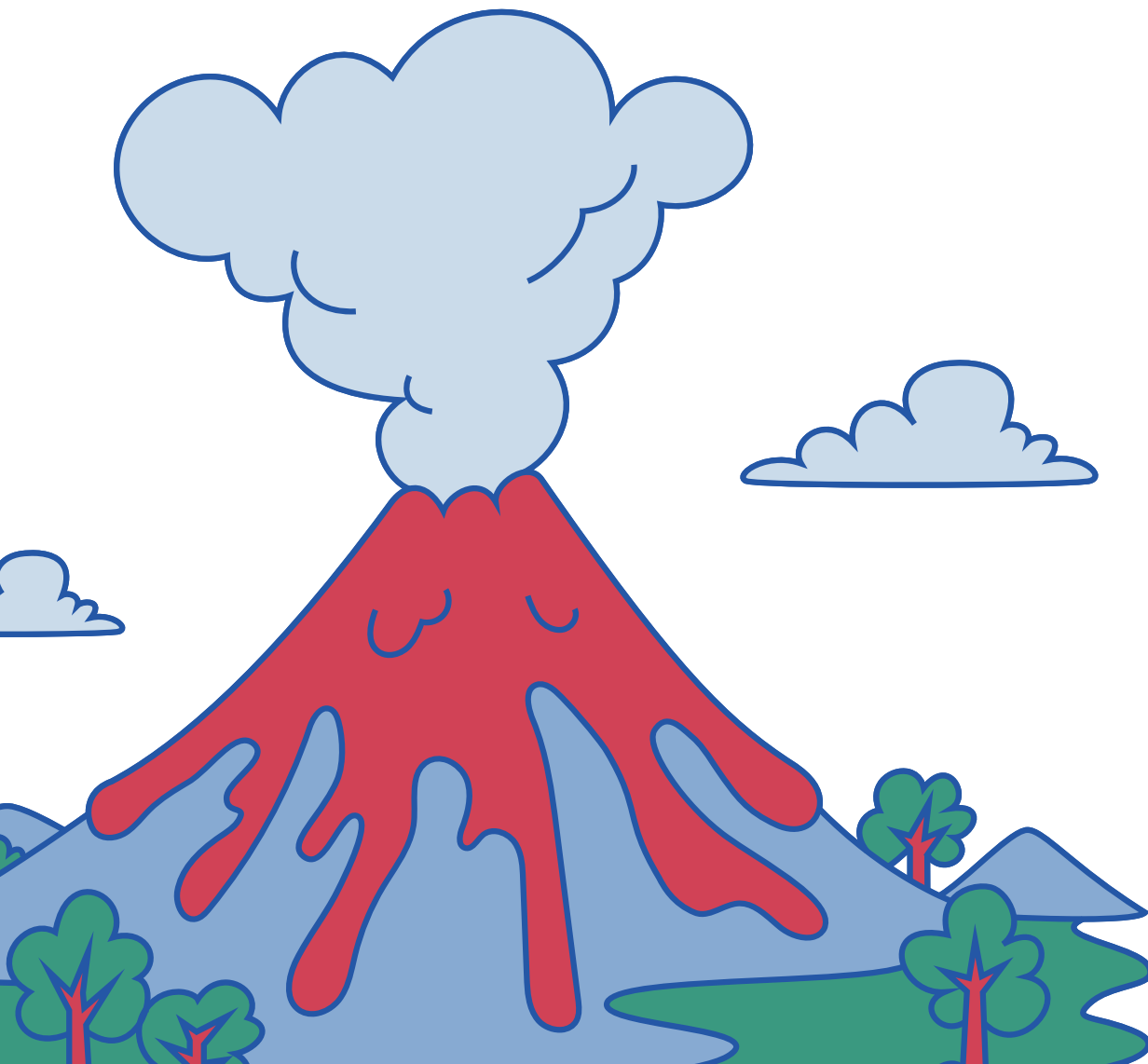
Ayusha K.C.

Prijina Gurung

Binaya Raj Shrestha

Bilakshana Neupane

Rahul Rimal



Introduction



Problem Statement

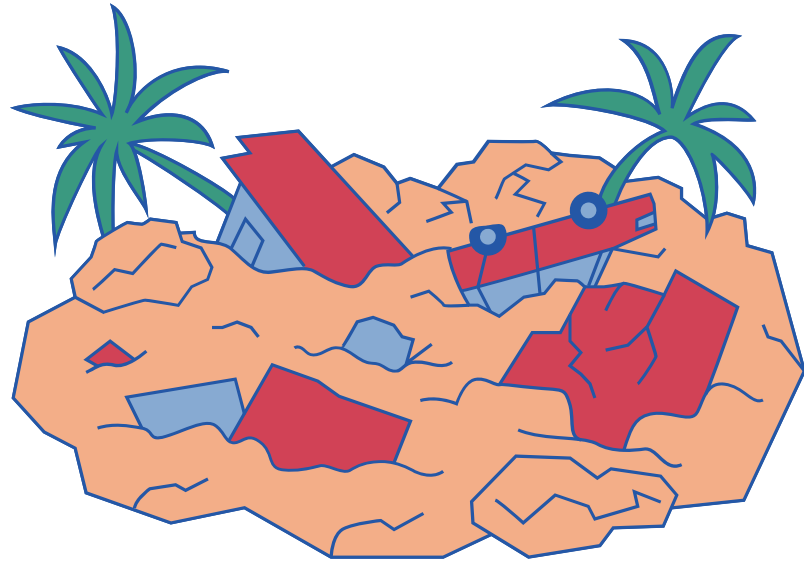
- Nepal frequently experiences floods, landslides, storms, and heatwaves.
- Emergency response and relief distribution can be delayed.
- Managing relief resources efficiently is a major challenge.

Project Objective

- Predict disaster risks using real-time weather data.
- Identify affected districts and estimate relief needs.
- Support resource allocation through an interactive dashboard.



Literature Review



Rapid Dashboard Development

(Dabral et al., 2021)

- Streamlit cuts dev time vs. Flask/Django (no separate front-end needed)
- Ideal for small student teams building fast, functional dashboards



Live Weather Monitoring

(Maurya & Singh, 2026)

- Python + OpenWeatherMap API reliably deliver accurate real-time data
- Validated against official weather benchmarks, low latency

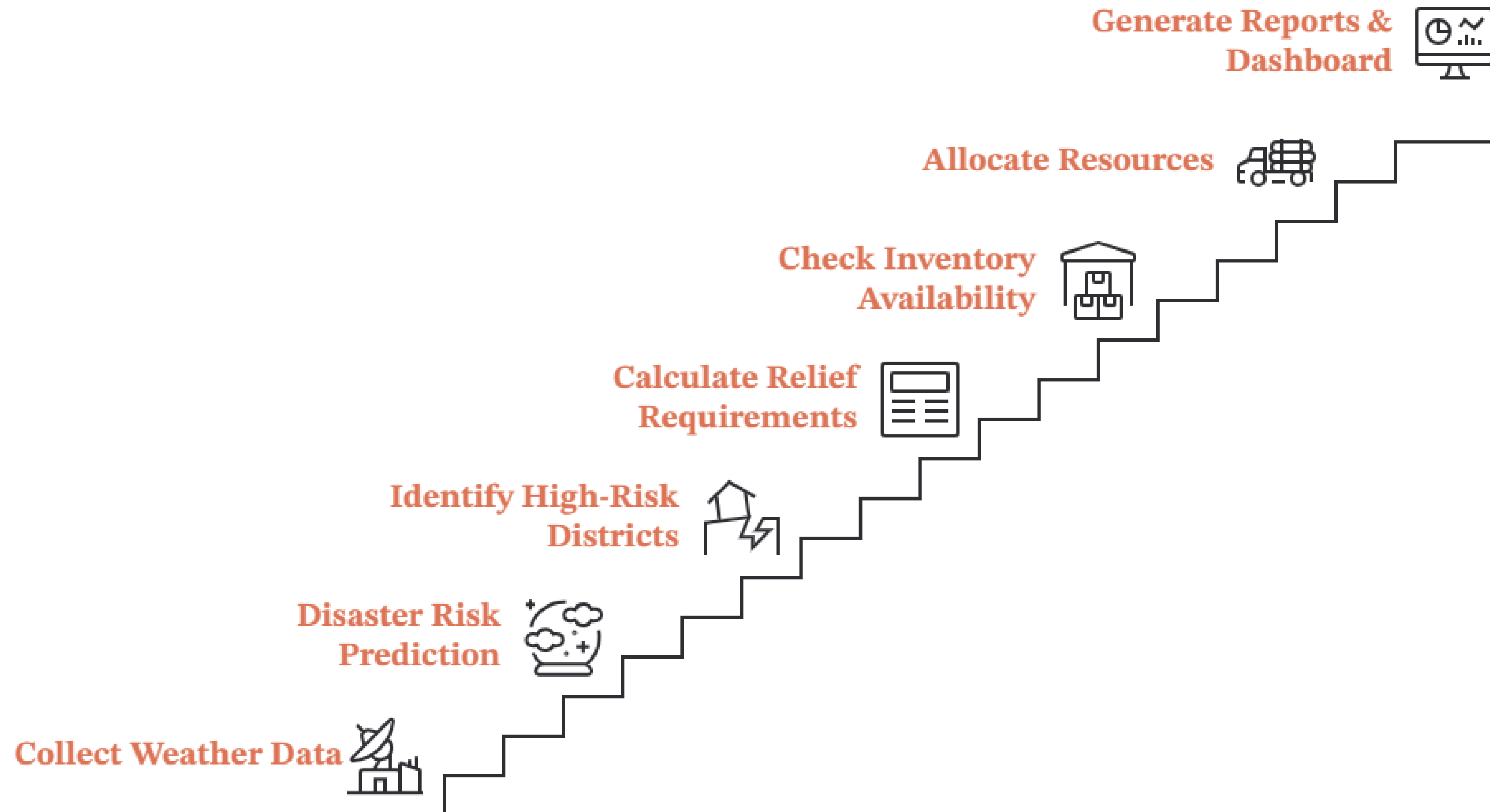


Real-Time Disaster Prediction

(Chaitanya et al., 2026)

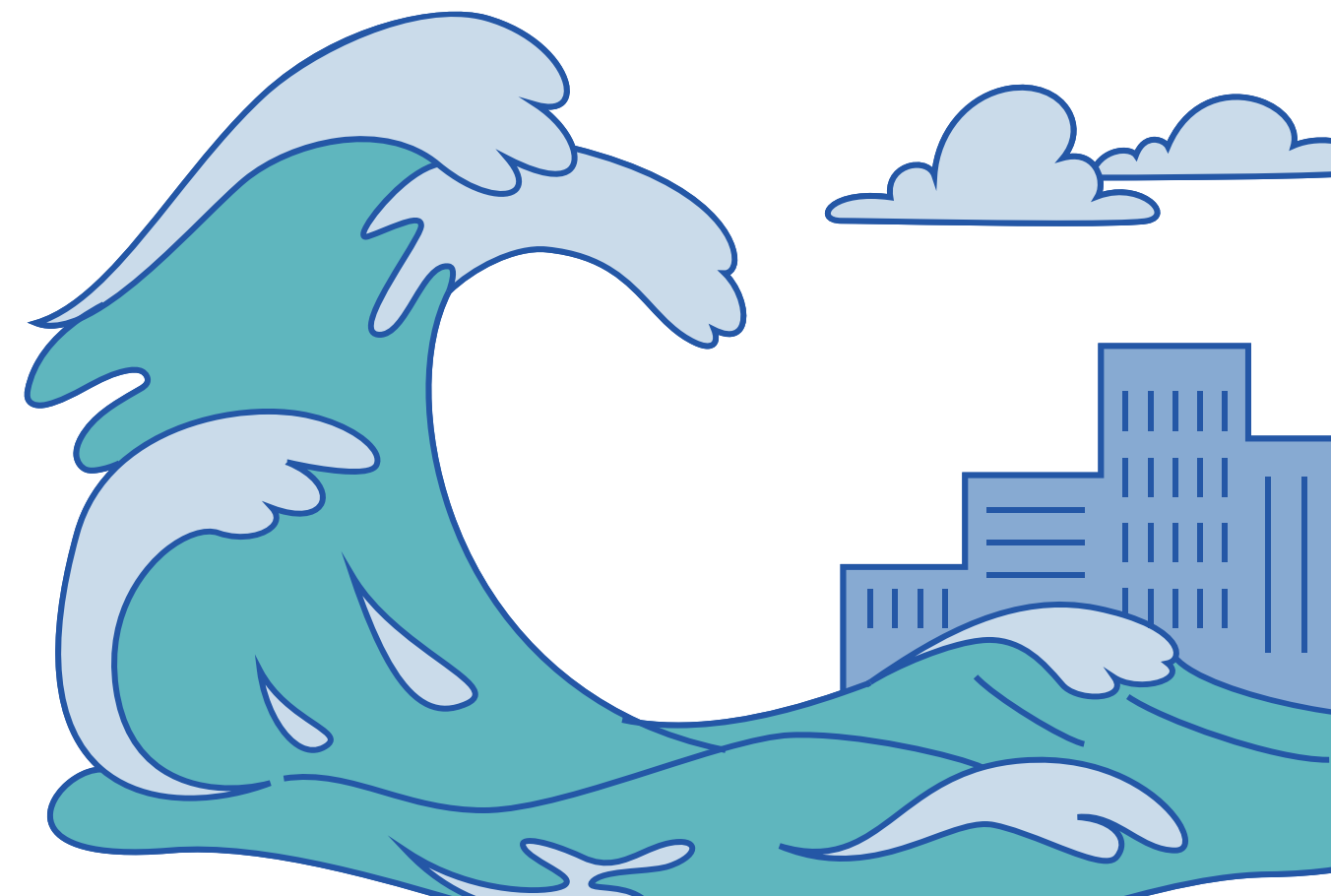
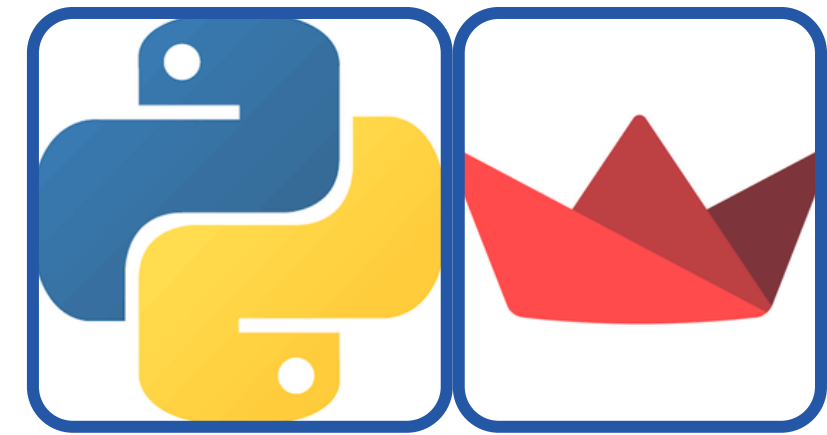
- ML + live dashboard achieved 94.6% accuracy in flood risk classification
- Proves weather-API + dashboard pattern works for disaster alerts

Methodology



Tools & Technologies

- Python
- Streamlit
- MySQL
- OpenWeatherMap API
- Pandas
- Requests
- Plotly



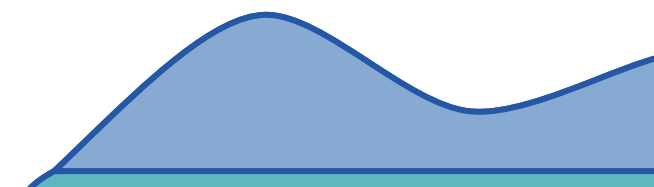
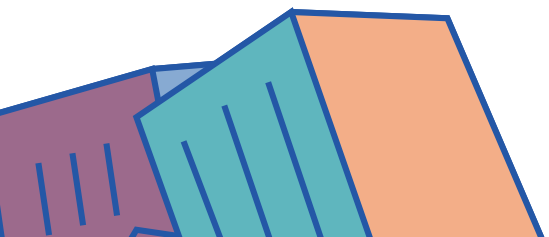
Expected Output

- Live weather monitoring
- Disaster risk prediction
- High-risk district identification
- Relief inventory management
- Automatic resource allocation
- Interactive dashboard and reports



Weekly Deliverables

Name	Start date	End date	August 2026																																										
			19	20	21	22	23	24	25	26	27	28	29	30	31	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
			S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S
Smart Disaster Prediction and Relief Management System	7/20/26	8/19/26																																											
▼ Planning & Design	7/20/26	7/26/26	▼ Planning & Design																																										
Requirements & System Design	7/20/26	7/23/26	Requirements & System Design																																										
Database Design	7/22/26	7/26/26	Database Design																																										
▼ Weather & Risk Prediction	7/27/26	8/9/26	▼ Weather & Risk Prediction																																										
Weather API Integration	7/27/26	8/2/26	Weather API Integration																																										
↳ Disaster Risk Prediction	7/30/26	8/9/26	↳ Disaster Risk Prediction																																										
▼ District & Relief Management	8/3/26	8/9/26	▼ District & Relief Man...																																										
District Vulnerability Database	8/3/26	8/6/26	District V...																																										
Relief Inventory Management	8/5/26	8/9/26	Relief Invento...																																										
▼ Resource Allocation & Dashboard	8/10/26	8/16/26	▼ Resource Allocation ...																																										
Resource Allocation & Recommendations	8/10/26	8/13/26	Resource ...																																										
Streamlit Dashboard	8/11/26	8/16/26	Streamlit Dashb...																																										
▼ Testing & Finalization	8/14/26	8/19/26	▼ Testing & Finaliz...																																										
System Integration & Testing	8/14/26	8/18/26	System Integ...																																										
Documentation & Final Presentation	8/17/26	8/19/26	Docu...																																										



References

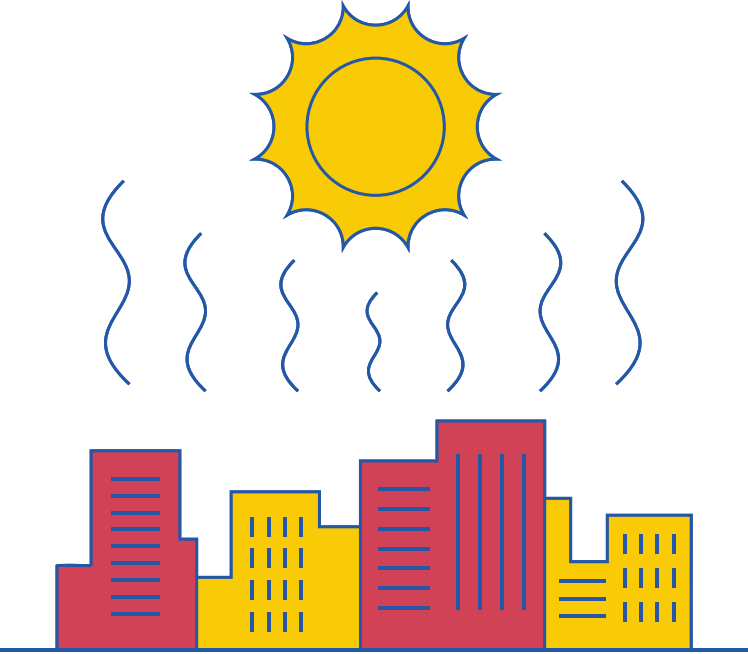
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- Maurya, S., & Singh, P. (2026). GUI-based weather dashboard using Python and OpenWeatherMap API. International Journal of Science, Engineering and Technology.
<https://doi.org/10.5281/zenodo.19761598>





Thank You





Data Analysis

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